16 Two stars are at a distance of 20 light years from the Earth. An optical telescope with an aperture of 3.0 m is used to view the stars.

A light year is the distance light travels in a vacuum in one year which is approximately 10^{16} m.

What is the order of magnitude of the minimum separation between the two stars that can just be resolved by the telescope?

A 10^{8} m **B** 10^{10} m **C** 10^{12} m **D** 10^{14} m